
Calculus BC

Practice Exam

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A A

CALCULUS BC

SECTION I, Part A

Time—1 hour

Number of questions—30

NO CALCULATOR IS ALLOWED FOR THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in this exam booklet. Do not spend too much time on any one problem.

In this exam:

- (1) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.
- (2) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix “arc” (e.g., $\sin^{-1} x = \arcsin x$).



1. If $f(x) = \cos^2(3x - 5)$, then $f'(x) =$
- (A) $6 \cos(3x - 5)$
- (B) $-3 \sin^2(3x - 5)$
- (C) $-2 \sin(3x - 5) \cos(3x - 5)$
- (D) $-6 \sin(3x - 5) \cos(3x - 5)$

2. $\int \frac{1}{t\sqrt{t}} dt =$

- (A) $-2t^{-1/2} + C$
 (B) $-\frac{3}{2}t^{-5/2} + C$
 (C) $-\frac{2}{5}t^{-5/2} + C$
 (D) $2t^{1/2}\ln t + C$

$$(A) \quad \frac{-4x^3 + 15x^2 - 2}{(x^3 + 2)^2}$$

(B) $\frac{-2x^3 + 15x^2 + 2}{(x^3 + 2)^2}$

(C) $\frac{2x^3 - 15x^2 - 2}{(x^3 + 2)^2}$

(D) $\frac{4x^3 - 15x^2 + 2}{(x^3 + 2)^2}$

22. $\int \frac{x}{2} e^{-3x/4} dx =$

$$(A) \quad -\frac{3x}{4}e^{-3x/4} + \frac{3}{4}e^{-3x/4} + C$$

(B) $-\frac{2x}{3}e^{-3x/4} - \frac{8}{9}e^{-3x/4} + C$

(C) $-\frac{x}{2}e^{-3x/4} + \frac{3}{8}e^{-3x/4} + C$

(D) $\frac{x}{2}e^{-3x/4} - \frac{1}{2}e^{-3x/4} + C$

27. The number of students in a cafeteria is modeled by the function P that satisfies the logistic differential equation $\frac{dP}{dt} = \frac{1}{2000} P(200 - P)$, where t is the time in seconds and $P(0) = 25$. What is the greatest rate of change, in students per second, of the number of students in the cafeteria?
- (A) 5 (B) 25 (C) 100 (D) 200

B**B****B****B****B****B****B****B****B****CALCULUS BC****SECTION I, Part B****Time—45 minutes****Number of questions—15**

A GRAPHING CALCULATOR IS REQUIRED FOR SOME QUESTIONS ON THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in this exam booklet. Do not spend too much time on any one problem.

BE SURE YOU ARE USING PAGE 3 OF THE ANSWER SHEET TO RECORD YOUR ANSWERS TO QUESTIONS NUMBERED 76–90.

YOU MAY NOT RETURN TO PAGE 2 OF THE ANSWER SHEET.

In this exam:

- (1) The exact numerical value of the correct answer does not always appear among the choices given. When this happens, select from among the choices the number that best approximates the exact numerical value.
- (2) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.
- (3) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix “arc” (e.g., $\sin^{-1} x = \arcsin x$).

B**B****B****B****B****B****B****B****B**

76. Let f be a twice-differentiable function for all real numbers x . Which of the following additional properties guarantees that f has a relative minimum at $x = c$?

(A) $f'(c) = 0$

(B) $f'(c) = 0$ and $f''(c) < 0$

(C) $f'(c) = 0$ and $f''(c) > 0$

(D) $f'(x) > 0$ for $x < c$ and $f'(x) < 0$ for $x > c$

77. Let $H(x)$ be an antiderivative of $\frac{x^3 + \sin x}{x^2 + 2}$. If $H(5) = \pi$, then $H(2) =$

(A) -9.008

(B) -5.867

(C) 4.626

(D) 12.150

B**B****B****B****B****B****B****B****B**

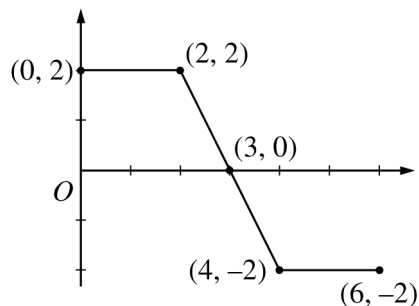
78. The continuous function f is positive and has domain $x > 0$. If the asymptotes of the graph of f are $x = 0$ and $y = 2$, which of the following statements must be true?

(A) $\lim_{x \rightarrow 0^+} f(x) = \infty$ and $\lim_{x \rightarrow 2} f(x) = \infty$

(B) $\lim_{x \rightarrow 0^+} f(x) = 2$ and $\lim_{x \rightarrow \infty} f(x) = 0$

(C) $\lim_{x \rightarrow 0^+} f(x) = \infty$ and $\lim_{x \rightarrow \infty} f(x) = 2$

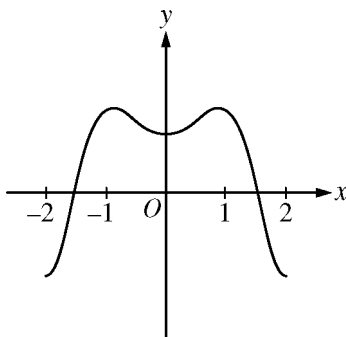
(D) $\lim_{x \rightarrow 2} f(x) = \infty$ and $\lim_{x \rightarrow \infty} f(x) = 2$

B**B****B****B****B****B****B****B****B**

Graph of f

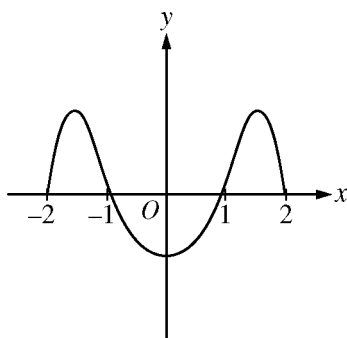
79. The graph of a function f , consisting of three line segments, is shown above. The function f is defined on the closed interval $[0, 6]$. Let $g(x) = \int_2^x f(t) dt$. What is the maximum value of $g(x)$ for $0 \leq x \leq 6$?
- (A) 0 (B) 1 (C) 5 (D) 10

-
80. The position of an object moving along a path in the xy -plane is given by the parametric equations $x(t) = 5 \sin(\pi t)$ and $y(t) = (2t - 1)^2$. The speed of the particle at time $t = 0$ is
- (A) 3.422
(B) 11.708
(C) 15.580
(D) 16.209

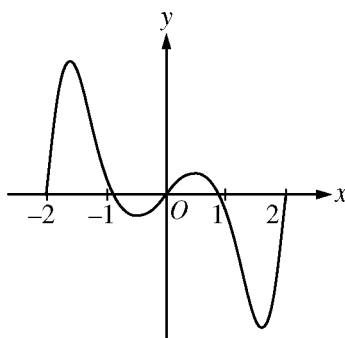
B**B****B****B****B****B****B****B****B**Graph of f

81. The graph of the function f is shown above for $-2 \leq x \leq 2$. Which of the following could be the graph of an antiderivative of f ?

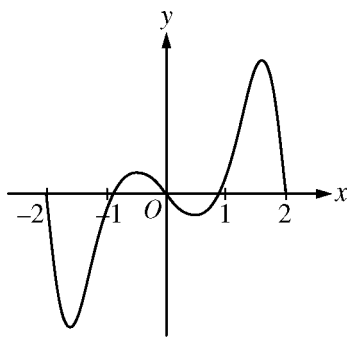
(A)



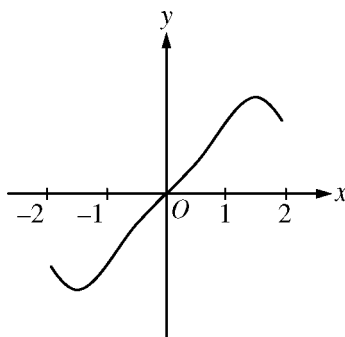
(B)



(C)



(D)



B**B****B****B****B****B****B****B****B**

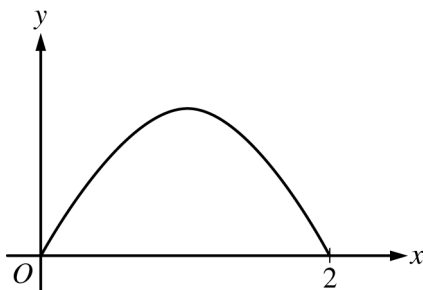
82. The derivative of the function f is given by $f'(x) = e^{-x}\cos(x^2)$, for all real numbers x . What is the minimum value of $f(x)$ for $-1 \leq x \leq 1$?

(A) $f(-1)$

(B) $f(-0.762)$

(C) $f(1)$

(D) There is no minimum value of $f(x)$ for $-1 \leq x \leq 1$.

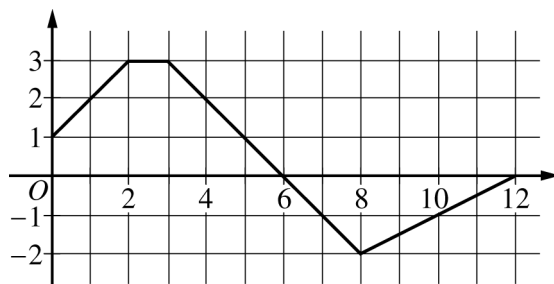
B**B****B****B****B****B****B****B****B**

83. The base of a solid is the region bounded by a portion of the graph of $y = \sin\left(\frac{\pi}{2}x\right)$ and the x -axis, as shown in the figure above. For the solid, each cross section perpendicular to the x -axis is a rectangle of height 3. Which of the following expressions gives the volume of the solid?

- (A) $\int_0^2 3 \sin\left(\frac{\pi}{2}x\right) dx$
- (B) $\int_0^2 3 \sin^2\left(\frac{\pi}{2}x\right) dx$
- (C) $\int_0^2 3\pi \sin\left(\frac{\pi}{2}x\right) dx$
- (D) $\int_0^2 3\pi \sin^2\left(\frac{\pi}{2}x\right) dx$

B**B****B****B****B****B****B****B****B**

84. If g is a twice-differentiable function, where $g(1) = 0.5$ and $\lim_{x \rightarrow \infty} g(x) = 4$, then $\int_1^{\infty} g'(x) \, dx$ is
- (A) -3.5 (B) 3.5 (C) 4.5 (D) nonexistent



Graph of f

85. The graph of the function f is shown above. If g is the function defined by $g(x) = \int_2^x f(t) \, dt$, what is the value of $g(10) \cdot g'(10)$?
- (A) $\frac{25}{4}$ (B) $\frac{5}{4}$ (C) $-\frac{5}{2}$ (D) $-\frac{25}{2}$

B**B****B****B****B****B****B****B****B**

$$f''(x) = x(x - 1)^2(x + 2)^3$$

$$g''(x) = x(x - 1)^2(x + 2)^3 + 1$$

$$h''(x) = x(x - 1)^2(x + 2)^3 - 1$$

86. The twice-differentiable functions f , g , and h have second derivatives given above. Which of the functions f , g , and h have a graph with exactly two points of inflection?
- (A) g only
(B) h only
(C) f and g only
(D) f , g , and h

-
87. The velocity vector of a particle moving in the xy -plane has components given by $\frac{dx}{dt} = \sin(t^2)$ and $\frac{dy}{dt} = e^{\cos t}$. At time $t = 4$, the position of the particle is $(2, 1)$. What is the y -coordinate of the position vector at time $t = 3$?

- (A) 0.410 (B) 0.590 (C) 0.851 (D) 1.410

B**B****B****B****B****B****B****B****B**

88. The function f is increasing on the interval $[1, 3]$ and nowhere else. The first derivative of f , f' , is continuous for all real numbers. Which of the following could be a table of values for $f'(x)$?

(A)

x	$f'(x)$
0	-1
1	0
2	2
3	0
4	-2

(B)

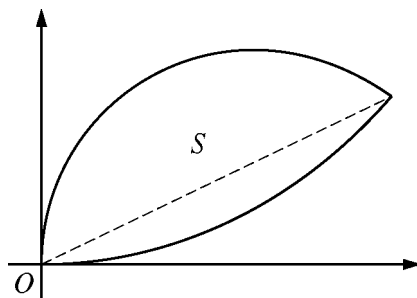
x	$f'(x)$
0	-1
1	1
2	2
3	1
4	-2

(C)

x	$f'(x)$
0	1
1	0
2	1
3	2
4	0

(D)

x	$f'(x)$
0	1
1	0
2	2
3	0
4	-2

B**B****B****B****B****B****B****B****B**

89. Let S be the region in the first quadrant bounded above by the graph of the polar curve $r = \cos \theta$ and bounded below by the graph of the polar curve $r = 2\theta$, as shown in the figure above. The two curves intersect when $\theta = 0.450$. What is the area of S ?

(A) 0.232 (B) 0.243 (C) 0.271 (D) 0.384

90. If the infinite series $S = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2}{n}$ is approximated by $P_k = \sum_{n=1}^k (-1)^{n+1} \frac{2}{n}$, what is the least value of k for which the alternating series error bound guarantees that $|S - P_k| < \frac{3}{100}$?

(A) 64 (B) 66 (C) 68 (D) 70

CALCULUS BC
SECTION II, Part A
Time—30 minutes
Number of questions—2

A GRAPHING CALCULATOR IS REQUIRED FOR THESE QUESTIONS.

1**1****1****1****1****1****1****1****1****1**

1. For $0 \leq t \leq 8$, a particle moving in the xy -plane has position vector $\langle x(t), y(t) \rangle = \langle \sin(2t), t^2 - t \rangle$, where $x(t)$ and $y(t)$ are measured in meters and t is measured in seconds.

(a) Find the speed of the particle at time $t = 2$ seconds. Indicate units of measure.

(b) At time $t = 4$ seconds, is the speed of the particle increasing or decreasing? Explain your answer.

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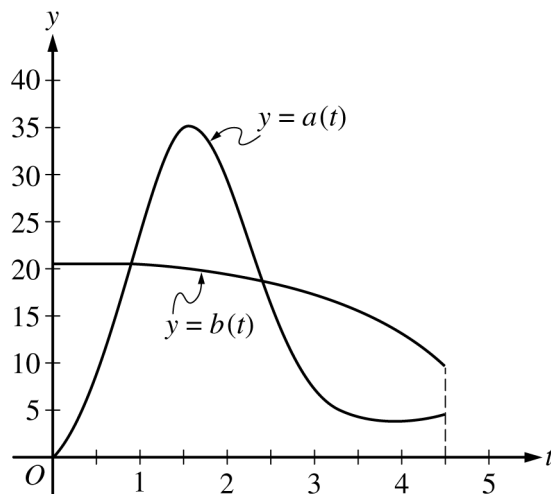
1**1****1****1****1****1****1****1****1****1**

(c) Find the total distance the particle travels over the time interval $0 \leq t \leq 5$ seconds.

(d) At time $t = 8$ seconds, the particle begins moving in a straight line. For $t \geq 8$, the particle travels with the same velocity vector that it had at time $t = 8$ seconds. Find the position of the particle at time $t = 10$ seconds.

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2. During the time interval $0 \leq t \leq 4.5$ hours, water flows into tank A at a rate of $a(t) = (2t - 5) + 5e^{2\sin t}$ liters per hour. During the same time interval, water flows into tank B at a rate of $b(t)$ liters per hour. Both tanks are empty at time $t = 0$. The graphs of $y = a(t)$ and $y = b(t)$, shown in the figure above, intersect at $t = k$ and $t = 2.416$.

(a) How much water will be in tank A at time $t = 4.5$?

- (b) During the time interval $0 \leq t \leq k$ hours, water flows into tank B at a constant rate of 20.5 liters per hour. What is the difference between the amount of water in tank A and the amount of water in tank B at time $t = k$?

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- (c) The area of the region bounded by the graphs of $y = a(t)$ and $y = b(t)$ for $k \leq t \leq 2.416$ is 14.470. How much water is in tank B at time $t = 2.416$?

-
- (d) During the time interval $2.7 \leq t \leq 4.5$ hours, the rate at which water flows into tank B is modeled by

$$w(t) = 21 - \frac{30t}{(t-8)^2} \text{ liters per hour. Is the difference } w(t) - a(t) \text{ increasing or decreasing at time}$$

$t = 3.5$? Show the work that leads to your answer.

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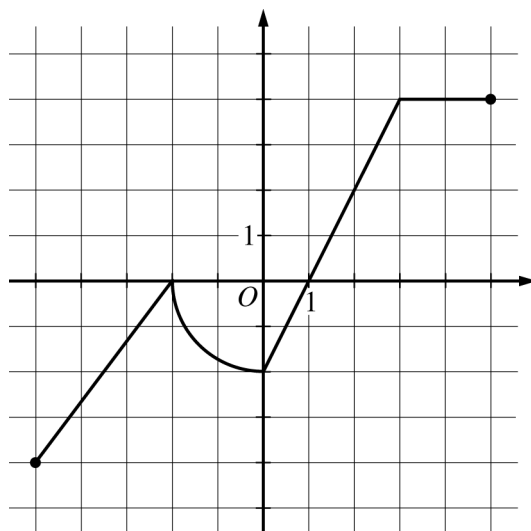
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CALCULUS BC
SECTION II, Part B
Time—1 hour
Number of questions—4

NO CALCULATOR IS ALLOWED FOR THESE QUESTIONS.

DO NOT BREAK THE SEALS UNTIL YOU ARE TOLD TO DO SO.

NO CALCULATOR ALLOWED

Graph of f

3. The graph of the function f , consisting of three line segments and a quarter of a circle, is shown above. Let g be the function defined by $g(x) = \int_1^x f(t) dt$.

(a) Find the average rate of change of g from $x = -5$ to $x = 5$.

(b) Find the instantaneous rate of change of g with respect to x at $x = 3$, or state that it does not exist.

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NO CALCULATOR ALLOWED

(c) On what open intervals, if any, is the graph of g concave up? Justify your answer.

(d) Find all x -values in the interval $-5 < x < 5$ at which g has a critical point. Classify each critical point as the location of a local minimum, a local maximum, or neither. Justify your answers.

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NO CALCULATOR ALLOWED

x	0	1	2	3	4	5	6
$f'(x)$	4	3.5	2	0.8	1.7	5.8	7

4. The function f satisfies $f(0) = 20$. The first derivative of f satisfies the inequality $0 \leq f'(x) \leq 7$ for all x in the closed interval $[0, 6]$. Selected values of f' are shown in the table above. The function f has a continuous second derivative for all real numbers.
- (a) Use a midpoint Riemann sum with three subintervals of equal length indicated by the data in the table to approximate the value of $f(6)$.

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- (b) Determine whether the actual value of $f(6)$ could be 70. Explain your reasoning.

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NO CALCULATOR ALLOWED

(c) Evaluate $\int_2^4 f''(x) \, dx$.

(d) Find $\lim_{x \rightarrow 0} \frac{f(x) - 20e^x}{0.5f(x) - 10}$.

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NO CALCULATOR ALLOWED

5. Consider the differential equation $\frac{dy}{dx} = -1 + \frac{y^2}{x}$.

(a) Show that $\frac{d^2y}{dx^2} = \frac{2y^3 - y^2 - 2xy}{x^2}$.

(b) Let $y = g(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -1 + \frac{y^2}{x}$ with initial condition

$g(4) = 2$. Does g have a relative minimum, a relative maximum, or neither at $x = 4$? Justify your

answer.

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NO CALCULATOR ALLOWED

- (c) Let $y = h(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -1 + \frac{y^2}{x}$ with initial condition $h(1) = 2$. Write the second-degree Taylor polynomial for h about $x = 1$.

-
- (d) For the function h given in part (c), it is known that $|h'''(x)| \leq 60$ for all x in the interval $0.9 \leq x \leq 1.1$. Let A represent the approximation of $h(1.1)$ found by using the second-degree Taylor polynomial for h about $x = 1$ from part (c). Use the Lagrange error bound to show that A differs from $h(1.1)$ by at most 0.01.

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NO CALCULATOR ALLOWED

6. Let f be the function defined by $f(x) = \frac{1}{x^2 + 9}$.

(a) Evaluate the improper integral $\int_3^{\infty} f(x) \, dx$, or show that the integral diverges.

(b) Determine whether the series $\sum_{n=3}^{\infty} f(n)$ converges or diverges. State the conditions of the test used for determining convergence or divergence.

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NO CALCULATOR ALLOWED

(c) Determine whether the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{(e^n \cdot f(n))} = \sum_{n=1}^{\infty} \frac{(-1)^n (n^2 + 9)}{e^n}$ converges absolutely, converges

conditionally, or diverges.

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Answer Key for AP Calculus BC Practice Exam, Section I

Question 1: D	Question 24: C
Question 2: A	Question 25: B
Question 3: C	Question 26: C
Question 4: D	Question 27: A
Question 5: C	Question 28: A
Question 6: C	Question 29: A
Question 7: D	Question 30: D
Question 8: D	Question 76: C
Question 9: B	Question 77: B
Question 10: C	Question 78: C
Question 11: A	Question 79: B
Question 12: D	Question 80: D
Question 13: B	Question 81: D
Question 14: C	Question 82: A
Question 15: C	Question 83: A
Question 16: A	Question 84: B
Question 17: B	Question 85: C
Question 18: B	Question 86: C
Question 19: C	Question 87: B
Question 20: B	Question 88: A
Question 21: B	Question 89: B
Question 22: B	Question 90: B
Question 23: D	

2017 AP Calculus BC

Question Descriptors and Performance Data

Multiple-Choice Questions

Question	Learning Objective	Essential Knowledge	Mathematical Practice for AP Calculus 1	Mathematical Practice for AP Calculus 2	Key	% Correct
1	2.1C	2.1C4	Implementing algebraic/computational processes	Building notational fluency	D	89
2	3.3B(a)	3.3B3	Implementing algebraic/computational processes	Building notational fluency	A	81
3	2.1C	2.1C3	Implementing algebraic/computational processes	Building notational fluency	C	88
4	2.3C	2.3C4	Implementing algebraic/computational processes	Connecting concepts	D	85
5	4.1B	4.1B1	Reasoning with definitions and theorems	Building notational fluency	C	58
6	1.2A	1.2A1	Connecting multiple representations	Reasoning with definitions and theorems	C	87
7	3.2C	3.2C2	Reasoning with definitions and theorems	Building notational fluency	D	84
8	3.4D	3.4D3	Reasoning with definitions and theorems	Connecting concepts	D	86
9	2.3B	2.3B1	Implementing algebraic/computational processes	Connecting concepts	B	90
10	2.3F	2.3F2	Implementing algebraic/computational processes	Building notational fluency	C	87
11	2.1C	2.1C5	Implementing algebraic/computational processes	Building notational fluency	A	83
12	3.3B(a)	3.3B5	Implementing algebraic/computational processes	Building notational fluency	D	81
13	2.3F	2.3F1	Connecting multiple representations	Connecting concepts	B	81
14	2.1A	2.1A3	Building notational fluency	Implementing algebraic/computational processes	C	83
15	1.1A(b)	1.1A3	Connecting multiple representations	Connecting concepts	C	54
16	4.2C	4.2C2	Implementing algebraic/computational processes	Connecting concepts	A	60
17	3.3B(b)	3.3B5	Implementing algebraic/computational processes	Building notational fluency	B	61
18	1.1A(b)	1.1A2	Connecting concepts	Implementing algebraic/computational processes	B	80
19	2.3B	2.3B2	Implementing algebraic/computational processes	Connecting concepts	C	69
20	2.1C	2.1C7	Implementing algebraic/computational processes	Connecting concepts	B	77
21	3.2B	3.2B2	Connecting multiple representations	Implementing algebraic/computational processes	B	71
22	3.3B(a)	3.3B5	Implementing algebraic/computational processes	Building notational fluency	B	66
23	4.2B	4.2B5	Implementing algebraic/computational processes	Building notational fluency	D	79
24	3.4B	3.4B1	Connecting concepts	Implementing algebraic/computational processes	C	64
25	4.2C	4.2C1	Connecting concepts	Reasoning with definitions and theorems	B	36
26	2.4A	2.4A1	Reasoning with definitions and theorems	Connecting concepts	C	50
27	3.5B	3.5B2	Implementing algebraic/computational processes	Connecting concepts	A	32
28	2.3C	2.3C2	Connecting concepts	Implementing algebraic/computational processes	A	80

2017 AP Calculus BC

Question Descriptors and Performance Data

Question	Learning Objective	Essential Knowledge	Mathematical Practice for AP Calculus 1	Mathematical Practice for AP Calculus 2	Key	% Correct
29	4.2B	4.2B2	Implementing algebraic/computational processes	Connecting concepts	A	53
30	4.1A	4.1A6	Reasoning with definitions and theorems	Building notational fluency	D	64
76	2.2A	2.2A1	Reasoning with definitions and theorems	Connecting concepts	C	76
77	3.3B(b)	3.3B2	Implementing algebraic/computational processes	Reasoning with definitions and theorems	B	74
78	1.1D	1.1D1	Building notational fluency	Connecting concepts	C	81
79	3.3A	3.3A3	Connecting multiple representations	Connecting concepts	B	67
80	2.3C	2.3C4	Implementing algebraic/computational processes	Connecting concepts	D	77
81	2.2A	2.2A3	Connecting multiple representations	Connecting concepts	D	65
82	2.2A	2.2A1	Implementing algebraic/computational processes	Connecting concepts	A	41
83	3.4D	3.4D2	Connecting concepts	Connecting multiple representations	A	67
84	3.2D	3.2D2	Connecting concepts	Building notational fluency	B	82
85	3.3A	3.3A3	Connecting multiple representations	Implementing algebraic/computational processes	C	74
86	2.2A	2.2A1	Connecting concepts	Connecting multiple representations	C	48
87	3.4C	3.4C2	Implementing algebraic/computational processes	Reasoning with definitions and theorems	B	65
88	2.2A	2.2A2	Connecting multiple representations	Connecting concepts	A	27
89	3.4D	3.4D1	Implementing algebraic/computational processes	Connecting multiple representations	B	37
90	4.1B	4.1B2	Reasoning with definitions and theorems	Connecting concepts	B	38

Free-Response Questions

Question	Learning Objective	Essential Knowledge	Mathematical Practice for AP Calculus	Mean
1	2.1C 2.2A 2.3C 3.4C	2.1C7 2.2A1 2.3C4 3.4C2	Reasoning with definitions and theorems Connecting concepts Implementing algebraic/computational processes Building notational fluency Communicating	3.9
2	2.3D 3.2C 3.4A 3.4D	2.3D1 3.2C2 3.4A2 3.4D1	Reasoning with definitions and theorems Connecting concepts Implementing algebraic/computational processes Connecting multiple representations Building notational fluency Communicating	4.94
3	2.1A 2.2A 3.2C 3.3A	2.1A1 2.2A1 3.2C1 3.3A2,3.3A3	Reasoning with definitions and theorems Connecting concepts Implementing algebraic/computational processes Connecting multiple representations Building notational fluency Communicating	5.15
4	1.1C 2.1C 3.2B 3.3B(b)	1.1C3 2.1C2 3.2B2 3.3B2	Reasoning with definitions and theorems Connecting concepts Implementing algebraic/computational processes Connecting multiple representations Building notational fluency Communicating	4.1
5	2.1C 2.1D 2.2A 4.2A	2.1C5 2.1D1 2.2A1 4.2A2,4.2A4	Reasoning with definitions and theorems Connecting concepts Implementing algebraic/computational processes Building notational fluency Communicating	4.81
6	3.2D 3.3B(b) 4.1A	3.2D2 3.3B5 4.1A4,4.1A6	Reasoning with definitions and theorems Connecting concepts Implementing algebraic/computational processes Building notational fluency Communicating	2.85

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Question 1

(a) $\sqrt{(x'(2))^2 + (y'(2))^2} = 3.272461$

The speed of the particle at time $t = 2$ seconds is 3.272 meters per second.

2 : $\begin{cases} 1 : \text{expression for speed} \\ 1 : \text{answer with units} \end{cases}$

(b) $s(t) = \sqrt{(x'(t))^2 + (y'(t))^2} = \sqrt{(2\cos(2t))^2 + (2t - 1)^2}$

$s'(4) = 2.16265$

Since $s'(4) > 0$, the speed of the particle is increasing at time $t = 4$.

2 : $\begin{cases} 1 : \text{considers } s'(4) \\ 1 : \text{answer with reason} \end{cases}$

(c) $\int_0^5 \sqrt{(x'(t))^2 + (y'(t))^2} dt = 22.381767$

The total distance the particle travels over the time interval $0 \leq t \leq 5$ seconds is 22.382 (or 22.381) meters.

2 : $\begin{cases} 1 : \text{integral} \\ 1 : \text{answer} \end{cases}$

(d) $x(10) = x(8) + x'(8) \cdot 2 = \sin 16 + x'(8) \cdot 2 = -4.118541$

$y(10) = y(8) + y'(8) \cdot 2 = (8^2 - 8) + y'(8) \cdot 2 = 86$

The position of the particle at time $t = 10$ seconds is $(-4.119, 86)$ (or $(-4.118, 86)$).

3 : $\begin{cases} 1 : \text{uses position at } t = 8 \\ 1 : \text{uses velocity at } t = 8 \\ 1 : \text{position at } t = 10 \end{cases}$

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Question 2

(a) $\int_0^{4.5} a(t) \, dt = 66.532128$

At time $t = 4.5$, tank A contains 66.532 liters of water.

2 : $\begin{cases} 1 : \text{integral} \\ 1 : \text{answer} \end{cases}$

(b) $a(k) = 20.5 \Rightarrow k = 0.892040$

$$\int_0^k (20.5 - a(t)) \, dt = 10.599191$$

At time $t = k$, the difference in the amounts of water in the tanks is 10.599 liters.

3 : $\begin{cases} 1 : \text{sets } a(k) = 20.5 \\ 1 : \text{integral} \\ 1 : \text{answer} \end{cases}$

(c) $\int_0^{2.416} b(t) \, dt = \int_0^k b(t) \, dt + \int_k^{2.416} b(t) \, dt$

$$\int_0^k b(t) \, dt = 20.5 \cdot k = 18.286826$$

On $k < t < 2.416$, tank A receives $\int_k^{2.416} a(t) \, dt = 44.497051$ liters of water, which is 14.470 more liters of water than tank B .

Therefore, $\int_k^{2.416} b(t) \, dt = \int_k^{2.416} a(t) \, dt - 14.470 = 30.027051$.

$$\int_0^k b(t) \, dt + \int_k^{2.416} b(t) \, dt = 48.313876$$

At time $t = 2.416$, tank B contains 48.314 (or 48.313) liters of water.

2 : $\begin{cases} 1 : \int_k^{2.416} a(t) \, dt \\ 1 : \text{answer} \end{cases}$

(d) $w'(3.5) - a'(3.5) = -1.14298 < 0$

The difference $w(t) - a(t)$ is decreasing at $t = 3.5$.

2 : $\begin{cases} 1 : w'(3.5) - a'(3.5) < 0 \\ 1 : \text{conclusion} \end{cases}$

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Question 3

(a) $\frac{g(5) - g(-5)}{5 - (-5)} = \frac{12 - (\pi + 7)}{10} = \frac{5 - \pi}{10}$

3 : $\begin{cases} 1 : \text{difference quotient} \\ 2 : \text{answer} \end{cases}$

(b) $g'(x) = f(x)$
 $g'(3) = f(3) = 4$

1 : answer

The instantaneous rate of change of g at $x = 3$ is 4.

(c) The graph of g is concave up on $-5 < x < -2$ and $0 < x < 3$,
because $g'(x) = f(x)$ is increasing on these intervals.

2 : intervals with justification

(d) $g'(x) = f(x)$ is defined at all x with $-5 < x < 5$.

$g'(x) = f(x) = 0$ at $x = -2$ and $x = 1$.

Therefore, g has critical points at $x = -2$ and $x = 1$.

g has neither a local maximum nor a local minimum at $x = -2$
because g' does not change sign there.

g has a local minimum at $x = 1$ because g' changes from negative to
positive there.

3 : $\begin{cases} 1 : \text{considers } f(x) = 0 \\ 1 : \text{critical points at} \\ \quad x = -2 \text{ and } x = 1 \\ 1 : \text{answers with justifications} \end{cases}$

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Question 4

(a) $\int_0^6 f'(x) \, dx \approx 2 \cdot 3.5 + 2 \cdot 0.8 + 2 \cdot 5.8 = 20.2$

$$f(6) - f(0) = \int_0^6 f'(x) \, dx$$

$$f(6) = f(0) + \int_0^6 f'(x) \, dx \approx 20 + 20.2 = 40.2$$

3 : $\begin{cases} 1 : \text{midpoint sum} \\ 1 : \text{Fundamental Theorem} \\ \quad \text{of Calculus} \\ 1 : \text{answer} \end{cases}$

(b) Since $f'(x) \leq 7$, $\int_0^6 f'(x) \, dx \leq 6 \cdot 7 = 42$.

$$f(6) - f(0) \leq 42 \Rightarrow f(6) \leq 20 + 42 = 62$$

Therefore, the actual value of $f(6)$ could not be 70.

2 : $\begin{cases} 1 : \text{integral bound} \\ 1 : \text{answer with reasoning} \end{cases}$

(c) $\int_2^4 f''(x) \, dx = f'(4) - f'(2) = 1.7 - 2 = -0.3$

2 : $\begin{cases} 1 : \text{Fundamental Theorem} \\ \quad \text{of Calculus} \\ 1 : \text{answer} \end{cases}$

(d) $\lim_{x \rightarrow 0} (f(x) - 20e^x) = 0$

$$\lim_{x \rightarrow 0} (0.5f(x) - 10) = 0$$

Using L'Hospital's Rule,

$$\lim_{x \rightarrow 0} \frac{f(x) - 20e^x}{0.5f(x) - 10} = \lim_{x \rightarrow 0} \frac{f'(x) - 20e^x}{0.5f'(x)} = \frac{4 - 20}{0.5(4)} = -8$$

2 : $\begin{cases} 1 : \text{L'Hospital's Rule} \\ 1 : \text{answer} \end{cases}$

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Question 5

$$\begin{aligned} \text{(a)} \quad \frac{d^2y}{dx^2} &= \frac{x \cdot 2y \frac{dy}{dx} - y^2 \cdot 1}{x^2} \\ &= \frac{2xy \left(-1 + \frac{y^2}{x} \right) - y^2}{x^2} = \frac{2y^3 - y^2 - 2xy}{x^2} \end{aligned}$$

$$2 : \begin{cases} 1 : \text{quotient rule} \\ 1 : \frac{d^2y}{dx^2} \end{cases}$$

$$\text{(b)} \quad \left. \frac{dy}{dx} \right|_{(x,y)=(4,2)} = -1 + \frac{4}{4} = 0$$

$$2 : \begin{cases} 1 : \text{considers } \left. \frac{dy}{dx} \right|_{(x,y)=(4,2)} \\ 1 : \text{answer with justification} \end{cases}$$

$$\left. \frac{d^2y}{dx^2} \right|_{(x,y)=(4,2)} = \frac{2 \cdot 8 - 4 - 16}{16} = -\frac{1}{4} < 0$$

By the Second Derivative Test, g has a relative maximum at $x = 4$.

$$\text{(c)} \quad \left. \frac{dy}{dx} \right|_{(x,y)=(1,2)} = -1 + \frac{4}{1} = 3$$

$$3 : \begin{cases} 1 : \text{uses } \left. \frac{dy}{dx} \right|_{(x,y)=(1,2)} \\ \text{and } \left. \frac{d^2y}{dx^2} \right|_{(x,y)=(1,2)} \\ 2 : \text{Taylor polynomial} \end{cases}$$

$$\left. \frac{d^2y}{dx^2} \right|_{(x,y)=(1,2)} = \frac{2 \cdot 8 - 4 - 4}{1} = 8$$

The second-degree Taylor polynomial for h about $x = 1$ is

$$T_2(x) = 2 + 3(x - 1) + \frac{8}{2!}(x - 1)^2 = 2 + 3(x - 1) + 4(x - 1)^2.$$

$$\text{(d)} \quad |h(1.1) - A| \leq \frac{\max_{1.0 \leq x \leq 1.1} |h'''(x)| |1.1 - 1|^3}{3!} \leq \frac{60}{6} \cdot \frac{1}{1000} = \frac{10}{1000} = \frac{1}{100}$$

$$2 : \begin{cases} 1 : \text{form of the error bound} \\ 1 : \text{analysis} \end{cases}$$

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Question 6

$$\begin{aligned} \text{(a)} \quad \int_3^\infty \frac{1}{x^2 + 9} dx &= \lim_{b \rightarrow \infty} \int_3^b \frac{1}{x^2 + 9} dx = \lim_{b \rightarrow \infty} \left(\frac{1}{3} \tan^{-1} \left(\frac{x}{3} \right) \Big|_3^b \right) \\ &= \lim_{b \rightarrow \infty} \left(\frac{1}{3} \tan^{-1} \left(\frac{b}{3} \right) - \frac{1}{3} \tan^{-1}(1) \right) = \frac{\pi}{6} - \frac{\pi}{12} = \frac{\pi}{12} \end{aligned}$$

3 : $\begin{cases} 1 : \text{antiderivative} \\ 1 : \text{limit expression} \\ 1 : \text{answer} \end{cases}$

(b) The function f is continuous, positive, and decreasing on $[3, \infty)$.

2 : conclusion with conditions

By the integral test, since $\int_3^\infty f(x) dx$ converges, $\sum_{n=3}^\infty f(n)$ converges.

— OR —

$$0 < \frac{1}{n^2 + 9} < \frac{1}{n^2} \text{ for } n \geq 3.$$

Since the series $\sum_{n=3}^\infty \frac{1}{n^2}$ converges, the series $\sum_{n=3}^\infty f(n) = \sum_{n=3}^\infty \frac{1}{n^2 + 9}$ converges by the comparison test.

$$\text{(c)} \quad \text{Consider the series } \sum_{n=1}^\infty \frac{1}{(e^n \cdot f(n))} = \sum_{n=1}^\infty \frac{n^2 + 9}{e^n}.$$

4 : $\begin{cases} 1 : \text{sets up ratio} \\ 1 : \text{computes limit of ratio} \\ 1 : \text{conclusion of ratio test} \\ 1 : \text{converges absolutely} \end{cases}$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{(n+1)^2 + 9}{e^{n+1}}}{\frac{n^2 + 9}{e^n}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)^2 + 9}{n^2 + 9} \cdot \frac{1}{e} \right| = \frac{1}{e} < 1$$

By the ratio test, $\sum_{n=1}^\infty \frac{1}{(e^n \cdot f(n))}$ converges.

Therefore, $\sum_{n=1}^\infty \frac{(-1)^n}{(e^n \cdot f(n))}$ converges absolutely.